

Selecting efficient pharmaceutical binder for developing robust extended-release matrices of Compritol® 888ATO prepared using wet granulation technique: Part 2 Studying impact on dissolution profiles of ER tablets of different API's

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OBJECTIVE:

This study evaluates impact of four pharmaceutical binders on dissolution profile of ER tablets of different API's formulated using Compritol® 888 ATO as sustained release agent.

METHODS:

Physicochemical Properties of binders under study

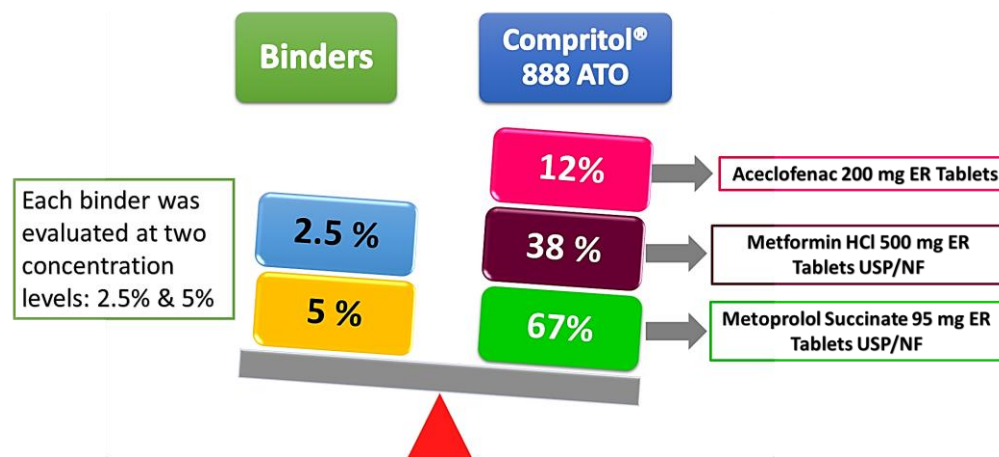
Excipients Properties	HPC L	HPMC E6	PVP K30	PVP K90
Chemical name	Hydroxy propyl cellulose	Hydroxy propyl methyl cellulose	poly(1-vinyl-2-pyrrolidone)	poly(1-vinyl-2-pyrrolidone)
Avg. molecular weight	95,000 Daltons	21,000 Daltons	54,000 Daltons	15,00,000 Daltons
Viscosity (5% solution at 25°C,	105 cps	15 cps	3 cps	150 cps
Melting point(°C)/Tg	Soften at 130°C	170-180°C	130°C	130°C
Appearance in Water at Room Temperature	Viscous Gel	Viscous Gel	Clear Solution	Clear Solution

Physicochemical Properties of APIs & Formulation Characteristics

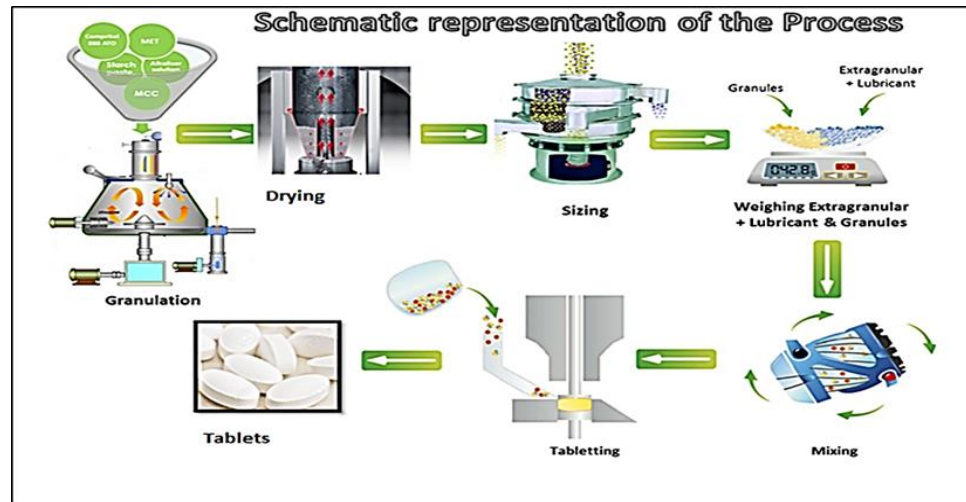
API Names	Aceclofenac	Metformin HCl	Metoprolol Succinate
Dose	200 mg	500 mg	95 mg
Log P	3.88	-0.92	1.76
Plasma half life	4 hours.	6.2 hours	3 to 7 hours.
Water solubility	Insoluble (pH dependent solubility)	> 50 mg/ml 200 mg/mL	Freely soluble
Concentration of Compritol® 888 ATO	12 %	38 %	67 %
Tablet weight	400 mg	1035 mg	850 mg
Tooling details	9.5 mm Round Standard Concave punch	19 X 9 mm Caplet punch	16 X 9 mm Caplet punch

Study Plan

ER tablet formulations for each API were prepared using following variations



Schematic representation of the Process



RESULTS:

Aceclofenac 200 mg ER Tablets

Compritol® 888 ATO: 12 %

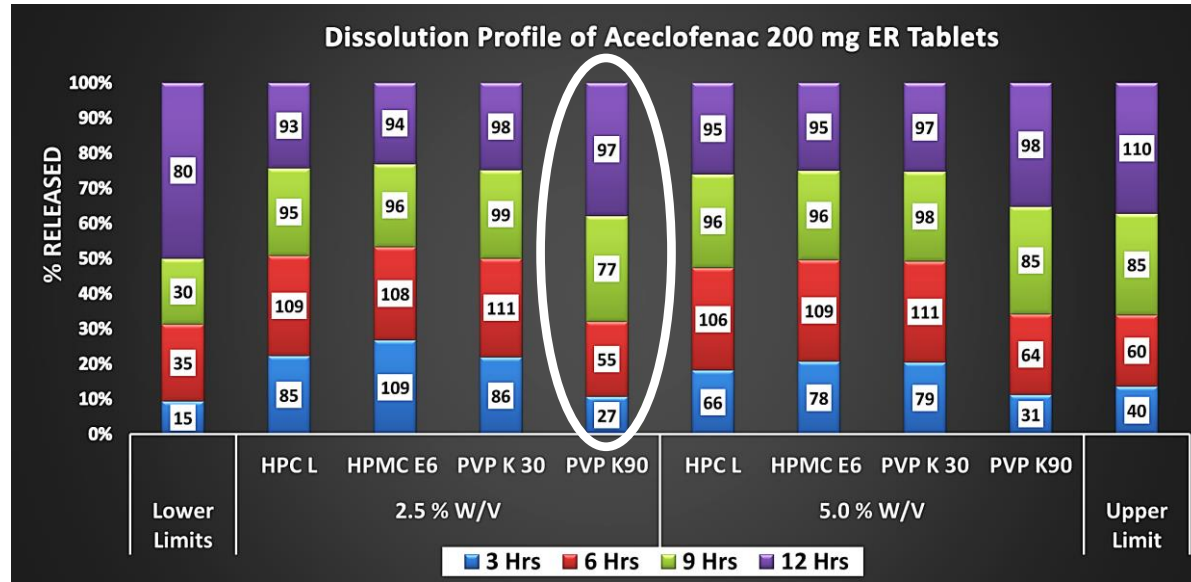


Fig. 1 Impact of Binders under study on Dissolution profile of Aceclofenac 200 mg ER Tablets

- Dissolution profile of the tablets prepared using PVP K90 at 2.5 % concentration was within limits of the desired profile.
- Dissolution of all other formulations failed to comply with the set limits.
- PVP K90 proved to be an efficient binder for forming a robust, sturdy SR tablet matrix of a poorly water-soluble API & waxy SR matrix forming agent such as Compritol® 888 ATO.

Metformin HCl 500mg ER Tablets USP/ NF

Compritol® 888 ATO: 38 %

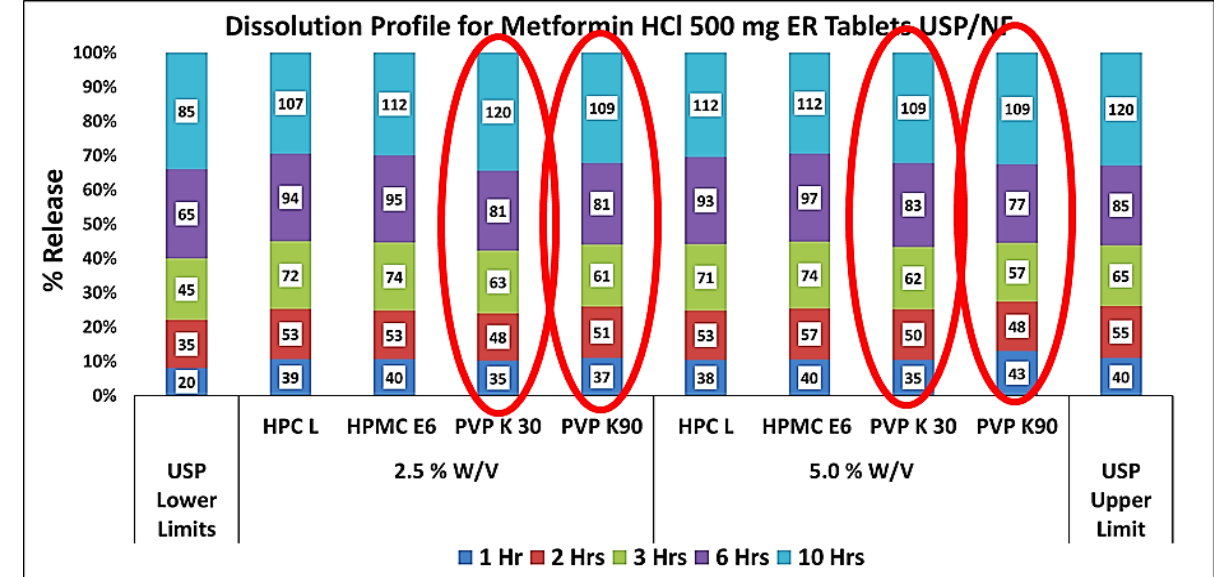


Fig. 2 Impact of Binders under study on Dissolution profile of Metformin HCl 500 mg ER Tablets USP/NF

- Dissolution profile of the tablets prepared using PVP K30 & PVP K90 at 2.5 % & 5 % concentration were found to be complying with Dissolution Test 1 & 2 of the USP/NF monograph on Metformin HCl ER Tablets.
- No impact of binder concentration on the release of Metformin HCl was observed for all the four binders under study.
- Thus, PVP K30 & PVP K90 proved to be efficient binders at 2.5% for providing a robust, sturdy SR tablet matrix of an API of high-water solubility (> 50 mg/ml) & high dose like Metformin HCl with waxy SR matrix forming agent such as Compritol® 888 ATO.

Metoprolol Succinate 95 mg ER Tablets USP

Compritol® 888 ATO: 67 %

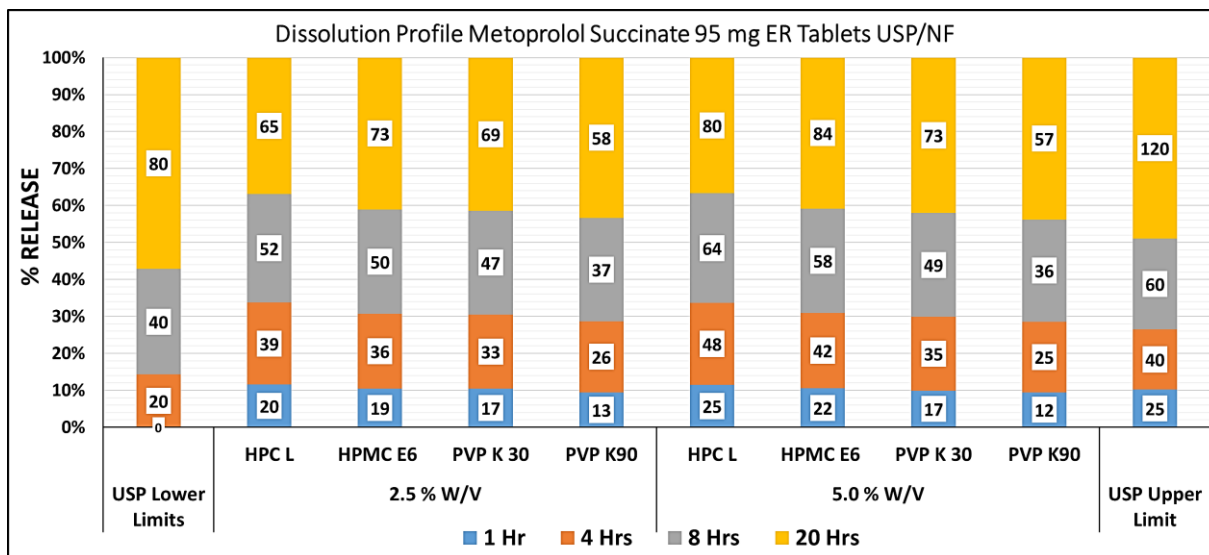


Fig. 3 Impact of Binders under study on Dissolution profile of Metoprolol Succinate 95 mg ER Tablets USP/NF

- Metoprolol Succinate is a freely water-soluble low dose API with 20 hours dissolution profile as per Dissolution Test 1 of USP/NF monograph on Metoprolol Succinate ER Tablets
- Water swellable binders such as HPC L & HPMC E6 at 2.5% concentration afforded profile that complied with required limits at borderline at all time points except at 20 hours where the release failed to comply with lower limits.
- Tablets prepared using PVP K90 & PVP K30 as binder at both 2.5% & 5% levels gave a controlled release profile that failed to comply with lower limit at 20 hours release (NLT 80 %).

CONCLUSIONS:

1. Our investigation demonstrates that binders with high viscosity & non swellable nature such as PVP K90 gives best results at lower concentrations for enhancing the binding between hydrophobic waxy Compritol® 888 ATO and hydrophilic API's.
2. Dissolution profiles afforded by PVP K90 used at lower concentration of 2.5% for all the three APIs under study complied with the requisite specifications & proved to be effective for all concentration variants of Compritol® 888 ATO (12 %, 38 % & 67 %).
3. PVP K30 was found to be effective only for highly water-soluble API's (Metformin HCl & Metoprolol Succinate).
4. **Thus, PVP K – 90 is recommended binder for producing Extended-Release Tablets containing Compritol® 888 ATO as release retarding agent at a wide concentration range (10 to 70 %) & for APIs with different solubilities.**

REFERENCES:

1. www.gattefosse.com
2. Morkhade DM. Comparative impact of different binder addition methods, binders and diluents on resulting granule and tablet attributes via high shear wet granulation. Powder technology. 2017 Oct 1; 320:114-120.